

Submit 8 problems total from the union of *LADW* Ch 5.1-3 & Ch 6.1-2 (copied below) and the supplementary problems.

Please label all problems. Please put a “G” in front of the four problems you would like graded for correctness. Additionally, put an “F” in front of any of the problems graded for completion on which you would like written feedback from the grader. Do not submit any of the problems marked with an X.

Suggested *LADW* problems: 5.1.5, 5.1.8, 5.2.1, 5.2.5, 5.3.5, 5.3.9, 5.3.11, 6.1.1, 6.2.1, 6.2.4, 6.2.8

Section 5.1

Exercises.

X 1.1. Compute $(3+2i)(5-3i)$, $\frac{2-3i}{1-2i}$, $\operatorname{Re}\left(\frac{2-3i}{1-2i}\right)$, $(1+2i)^3$, $\operatorname{Im}((1+2i)^3)$.

X 1.2. For vectors $\mathbf{x} = (1, 2i, 1+i)^T$ and $\mathbf{y} = (i, 2-i, 3)^T$ compute

- $\langle \mathbf{x}, \mathbf{y} \rangle$, $\|\mathbf{x}\|^2$, $\|\mathbf{y}\|^2$, $\|\mathbf{y}\|$;
- $(3\mathbf{x}, 2i\mathbf{y})$, $(2\mathbf{x}, i\mathbf{x} + 2\mathbf{y})$;
- $\|\mathbf{x} + 2\mathbf{y}\|$.

Remark: After you have done part a), you can do parts b) and c) without actually computing all vectors involved, just by using the properties of inner product.

X 1.3. Let $\|\mathbf{u}\| = 2$, $\|\mathbf{v}\| = 3$, $\langle \mathbf{u}, \mathbf{v} \rangle = 2 + i$. Compute $\|\mathbf{u} + \mathbf{v}\|^2$, $\|\mathbf{u} - \mathbf{v}\|^2$, $\langle \mathbf{u} + \mathbf{v}, \mathbf{u} - i\mathbf{v} \rangle$, $\langle \mathbf{u} + 3i\mathbf{v}, 4i\mathbf{u} \rangle$.

1.4. Prove that for vectors in an inner product space

$$\|\mathbf{x} \pm \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 \pm 2\operatorname{Re}\langle \mathbf{x}, \mathbf{y} \rangle$$

Recall that $\operatorname{Re} z = \frac{1}{2}(z + \bar{z})$

1.5. Explain why each of the following is *not* an inner product on a given vector space:

- $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 - x_2 y_2$ on $\mathbb{R}^2$;
- $\langle A, B \rangle = \operatorname{trace}(A + B)$ on the space of real 2×2 matrices;
- $\langle f, g \rangle = \int_0^1 f'(t)g(t)dt$ on the space of polynomials; $f'(t)$ denotes derivative.

1.6 (Equality in Cauchy–Schwarz inequality). Prove that

$$|\langle \mathbf{x}, \mathbf{y} \rangle| = \|\mathbf{x}\| \cdot \|\mathbf{y}\|$$

if and only if one of the vectors is a multiple of the other. **Hint:** Analyze the proof of the Cauchy–Schwarz inequality.

1.7. Prove the parallelogram identity for an inner product space V ,

$$\|\mathbf{x} + \mathbf{y}\|^2 + \|\mathbf{x} - \mathbf{y}\|^2 = 2(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2).$$

1.8. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be a spanning set (in particular, a basis) in an inner product space V . Prove that

- If $\langle \mathbf{x}, \mathbf{v} \rangle = 0$ for all $\mathbf{v} \in V$, then $\mathbf{x} = \mathbf{0}$;
- If $\langle \mathbf{x}, \mathbf{v}_k \rangle = 0 \ \forall k$, then $\mathbf{x} = \mathbf{0}$;
- If $\langle \mathbf{x}, \mathbf{v}_k \rangle = \langle \mathbf{y}, \mathbf{v}_k \rangle \ \forall k$, then $\mathbf{x} = \mathbf{y}$.

1.9. Consider the space $\mathbb{R}^2$ with the norm $\|\cdot\|_p$, introduced in Section 1.5. For $p = 1, 2, \infty$, draw the “unit ball” B_p in the norm $\|\cdot\|_p$

$$B_p := \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_p \leq 1\}.$$

Can you guess what the balls B_p for other p look like?

Section 5.2

Exercises.

2.1. Find all vectors in $\mathbb{R}^4$ orthogonal to vectors $(1, 1, 1, 1)^T$ and $(1, 2, 3, 4)^T$.

2.2. Let A be a real $m \times n$ matrix. Describe $(\operatorname{Ran} A^T)^\perp$, $(\operatorname{Ran} A)^\perp$

2.3. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be an orthonormal basis in V .

- Prove that for any $\mathbf{x} = \sum_{k=1}^n \alpha_k \mathbf{v}_k$, $\mathbf{y} = \sum_{k=1}^n \beta_k \mathbf{v}_k$

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{k=1}^n \alpha_k \bar{\beta}_k.$$

- Deduce from this the *Parseval’s identity*

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{k=1}^n \langle \mathbf{x}, \mathbf{v}_k \rangle \overline{\langle \mathbf{y}, \mathbf{v}_k \rangle}$$

- Assume now that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is only an orthogonal basis, not an orthonormal one. Can you write down Parseval’s identity in this case?

This problem shows that if we have an orthonormal basis, we can use the coordinates in this basis absolutely the same way we use the standard coordinates in $\mathbb{C}^n$ (or $\mathbb{R}^n$).

The problem below shows that we can define an inner product by declaring a basis to be an orthonormal one.

2.4. Let V be a vector space and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be a basis in V . For $\mathbf{x} = \sum_{k=1}^n \alpha_k \mathbf{v}_k$, $\mathbf{y} = \sum_{k=1}^n \beta_k \mathbf{v}_k$ define $\langle \mathbf{x}, \mathbf{y} \rangle := \sum_{k=1}^n \alpha_k \bar{\beta}_k$.

Prove that $\langle \mathbf{x}, \mathbf{y} \rangle$ defines an inner product in V .

2.5. Let A be a real $m \times n$ matrix. Describe the set of all vectors in $\mathbb{R}^m$ orthogonal to $\operatorname{Ran} A$.

Section 5.3

Exercises.

X 3.1. Apply Gram–Schmidt orthogonalization to the system of vectors $(1, 2, -2)^T$, $(1, -1, 4)^T$, $(2, 1, 1)^T$.

3.2. Apply Gram–Schmidt orthogonalization to the system of vectors $(1, 2, 3)^T$, $(1, 3, 1)^T$. Write the matrix of the orthogonal projection onto 2-dimensional subspace spanned by these vectors.

3.3. Complete an orthogonal system obtained in the previous problem to an orthogonal basis in $\mathbb{R}^3$, i.e. add to the system some vectors (how many?) to get an orthogonal basis.

Can you describe how to complete an orthogonal system to an orthogonal basis in general situation of $\mathbb{R}^n$ or $\mathbb{C}^n$?

3.4. Find the distance from a vector $(2, 3, 1)^T$ to the subspace spanned by the vectors $(1, 2, 3)^T$, $(1, 3, 1)^T$. Note, that I am only asking to find the distance to the subspace, not the orthogonal projection.

3.5. Find the orthogonal projection of a vector $(1, 1, 1, 1)^T$ onto the subspace spanned by the vectors $\mathbf{v}_1 = (1, 3, 1, 1)^T$ and $\mathbf{v}_2 = (2, -1, 1, 0)^T$ (note that $\mathbf{v}_1 \perp \mathbf{v}_2$).

3.6. Find the distance from a vector $(1, 2, 3, 4)^T$ to the subspace spanned by the vectors $\mathbf{v}_1 = (1, -1, 1, 0)^T$ and $\mathbf{v}_2 = (1, 2, 1, 1)^T$ (note that $\mathbf{v}_1 \perp \mathbf{v}_2$). Can you find the distance without actually computing the projection? That would simplify the calculations.

3.7. True or false: if E is a subspace of V , then $\dim E + \dim(E^\perp) = \dim V$? Justify.

3.8. Let P be the orthogonal projection onto a subspace E of an inner product space V , $\dim V = n$, $\dim E = r$. Find the eigenvalues and the eigenvectors (eigenspaces). Find the algebraic and geometric multiplicities of each eigenvalue.

3.9. (Using eigenvalues to compute determinants).

- Find the matrix of the orthogonal projection onto the one-dimensional subspace in $\mathbb{R}^n$ spanned by the vector $(1, 1, \dots, 1)^T$;
- Let A be the $n \times n$ matrix with all entries equal 1. Compute its eigenvalues and their multiplicities (use the previous problem);
- Compute eigenvalues (and multiplicities) of the matrix $A - I$, i.e. of the matrix with zeroes on the main diagonal and ones everywhere else;
- Compute $\det(A - I)$.

3.10 (Legendre’s polynomials:). Let an inner product on the space of polynomials be defined by $\langle f, g \rangle = \int_{-1}^1 f(t)g(t)dt$. Apply Gram–Schmidt orthogonalization to the system $1, t, t^2, t^3$.

Legendre’s polynomials are particular case of the so-called orthogonal polynomials, which play an important role in many branches of mathematics.

3.11. Let $P = P_E$ be the matrix of an orthogonal projection onto a subspace E . Show that

- The matrix P is *self-adjoint*, meaning that $P^* = P$.
- $P^2 = P$.

Remark: The above 2 properties completely characterize orthogonal projection, i.e. any matrix P satisfying these properties is the matrix of some orthogonal projection. We will discuss this some time later.

3.12. Show that for a subspace E we have $(E^\perp)^\perp = E$. **Hint:** It is easy to see that E is orthogonal to $E^\perp$ (why?). To show that *any* vector $\mathbf{x}$ orthogonal to $E^\perp$ belongs to E use the decomposition $V = E \oplus E^\perp$ from Section 3.3 above.

3.13. Suppose P is the orthogonal projection onto a subspace E , and Q is the orthogonal projection onto the orthogonal complement $E^\perp$.

- What are $P + Q$ and PQ ?
- Show that $P - Q$ is its own inverse.

Section 6.1

Exercises.

1.1. Use the upper triangular representation of an operator to give an alternative proof of the fact that determinant is the product and the trace is the sum of eigenvalues counting multiplicities.

Section 6.2

Exercises.

2.1. True or false:

- Every unitary operator $U : X \rightarrow X$ is normal.
- A matrix is unitary if and only if it is invertible.
- If two matrices are unitarily equivalent, then they are also similar.
- The sum of self-adjoint operators is self-adjoint.
- The adjoint of a unitary operator is unitary.
- The adjoint of a normal operator is normal.
- If all eigenvalues of a linear operator are 1, then the operator must be unitary or orthogonal.
- If all eigenvalues of a normal operator are 1, then the operator is identity.
- A linear operator may preserve norm but not the inner product.

2.2. True or false: The sum of normal operators is normal? Justify your conclusion.

2.3. Show that an operator unitarily equivalent to a diagonal one is normal.

2.4. Orthogonally diagonalize the matrix,

$$A = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}.$$

Find all square roots of A , i.e. find all matrices B such that $B^2 = A$.

Note, that all square roots of A are self-adjoint.

2.5. True or false: any self-adjoint matrix has a self-adjoint square root. Justify.

2.6. Orthogonally diagonalize the matrix,

$$A = \begin{pmatrix} 7 & 2 \\ 2 & 4 \end{pmatrix},$$

i.e. represent it as $A = UDU^*$, where D is diagonal and U is unitary.

Among all square roots of A , i.e. among all matrices B such that $B^2 = A$, find one that has positive eigenvalues. You can leave B as a product.

2.7. True or false:

- A product of two self-adjoint matrices is self-adjoint.
- If A is self-adjoint, then A^k is self-adjoint.

Justify your conclusions

2.8. Let A be $m \times n$ matrix. Prove that

- A^*A is self-adjoint.
- All eigenvalues of A^*A are non-negative.
- $A^*A + I$ is invertible.

2.9. Give a proof if the statement is true, or give a counterexample if it is false:

- If $A = A^*$ then $A + iI$ is invertible.
- If U is unitary, $U + \frac{3}{4}I$ is invertible.
- If a matrix A is real, $A - iI$ is invertible.

2.10. Orthogonally diagonalize the rotation matrix

$$R_\alpha = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix},$$

where α is not a multiple of π . Note, that you will get complex eigenvalues in this case.

2.11. Orthogonally diagonalize the matrix

$$A = \begin{pmatrix} \cos \alpha & \sin \alpha \\ \sin \alpha & -\cos \alpha \end{pmatrix}.$$

Hints: You will get real eigenvalues in this case. Also, the trigonometric identities $\sin 2x = 2 \sin x \cos x$, $\sin^2 x = (1 - \cos 2x)/2$, $\cos^2 x = (1 + \cos 2x)/2$ (applied to $x = \alpha/2$) will help to simplify expressions for eigenvectors.

2.12. Can you describe the linear transformation with matrix A from the previous problem geometrically? Is has a very simple geometric interpretation.

2.13. Prove that a normal operator with unimodular eigenvalues (i.e. with all eigenvalues satisfying $|\lambda_k| = 1$) is unitary. **Hint:** Consider diagonalization

2.14. Prove that a normal operator with real eigenvalues is self-adjoint.

2.15. Show by example that conclusion of Theorem 2.2 fails for *complex* symmetric matrices. Namely

- construct a (diagonalizable) 2×2 complex symmetric matrix not admitting an orthogonal basis of eigenvectors;
- construct a 2×2 complex symmetric matrix which cannot be diagonalized.

Supplementary problems for HW6

I. The standard dot product still makes sense in $(\mathbb{Z}/5\mathbb{Z})^2$. (Despite the fact that it's not clear what the "non-negativity" condition of an inner product would mean in this setting.) Call two vectors in $(\mathbb{Z}/5\mathbb{Z})^2$ orthogonal if their dot product is 0. How many vectors in $(\mathbb{Z}/5\mathbb{Z})^2$ are orthogonal to the vector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$? Does each non-zero vector in $(\mathbb{Z}/5\mathbb{Z})^2$ have the same number of vectors orthogonal to it, or not?

II. Always, sometimes, or never? Prove if always or never, give an example each way if sometimes.

i) A given non-zero vector in $\mathbb{R}^n$ can be part of an orthonormal basis for $\mathbb{R}^n$.

ii) The product of two orthogonal matrices is orthogonal.

iii) If two vectors u and v are orthogonal, then a given pair of subspaces U and V containing u and v respectively are also orthogonal.

III. Always, sometimes, or never?

i) The projection of u onto v has a larger Euclidean norm than u .

ii) The dot product of u and v is larger than the Euclidean norm of each of u and v .

iii) The projection of u onto v has the same Euclidean norm as the projection of v onto u .

IV. Always, sometimes, or never? (Maybe try thinking about $\mathbb{R}^2$ first.)

i) In $\mathbb{R}^3$, the product of two reflections across planes is a reflection across a plane.

ii) In $\mathbb{R}^3$, if A and B are each an orthogonal projection to a subspace (perhaps different ones), then AB is an orthogonal projection to a subspace.

iii) In $\mathbb{R}^3$, the product of two rotations about lines is a rotation about a line.

V. Show that for a collection A of linearly independent vectors in $\mathbb{R}^n$, there exists a vector $w \in \mathbb{R}^n$ such that $w \cdot a > 0$ for all $a \in A$. Maybe start by thinking about $\mathbb{R}^2$ and then $\mathbb{R}^3$.